# Christos Zarkos

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## Education

**Massachusetts Institute of Technology** (Cambridge, MA, USA) Sept. 2023 - May 2028 (expected)
PhD in Electrical Engineering and Computer Science, advised by Prof. Christina Delimitrou

**Massachusetts Institute of Technology** (Cambridge, MA, USA) Sept. 2023 - May 2025
Master's of Science in Electrical Engineering and Computer Science, advised by Prof. Christina Delimitrou

**University of Crete** (Heraklion, Greece) Sept. 2019 - June 2023
Bachelor's in Computer Science ***GPA: 9.62/10.0 (Valedictorian)***

**Universitat Polytecnic de Catalunya** (Barcelona, Spain) Sept. 2022 - Jan. 2023
Erasmus+ exchange program, studying Computer Science

## Experience

**Massachusetts Institute of Technology** Sept. 2026 - December 2026

***Teaching Assistant for the class 6.5900 Computer System Architecture***

- Holding recitations, weekly office hours and grading assignments, and quiz exams

**Tenstorrent** May 2026 - August 2026

***Intern***

- Working on understanding and characterizing data movement bottlenecks to inform next-gen hardware and profiling infrastructure design, mentored by Neeraj Kulkarni. Worked in Santa Clara, California.

**Microsoft Research Redmond** May 2024 - August 2024

***Research Intern***

- Worked on AI-driven Hardware Design, alongside Shuotao Xu and Lei Qu

**University of Crete** Feb. 2023 - June 2023

***Teaching Assistant for the class CS255 Systems Programming Lab***

- Held recitations, weekly office hours and graded assignments, midterm and final exams

**Barcelona Supercomputing Center** Sept. 2022 - Jan 2023

***Visiting Junior Researcher***

- Implemented part of my bachelor's thesis as a member of the CAOS group, advised by Dr. Leonidas Kosmidis

## Publications

**Giza: The Case for Uncertainty-Governed Predictor Hierarchies in ML for Systems** (under submission)

***Varun Gohil*, Christos Zarkos*, Nevena Stojkovic, Christina Delimitrou***

**SERenaDE: Hardware Acceleration of Serialization Frameworks** Master Thesis (MICRO, November 2026)

***Christos Zarkos, Qihang Chen, Nikita Lazarev, Shabnam Sheikhha, James Tsai, Andy Anderson, Bhusan Chitlur, Christina Delimitrou***

- Accelerating serialization through repurposing an existing on-chip accelerator (Intel IAA), and utilizing its compression capabilities in the serialization process
- Transparently improving serialization performance through hardware offload, without the addition of extra hadware

## Workshop Publications

**The Case for Uncertainty-Governed Predictor Hierarchies in ML for Systems** (Architecture 2.0)

***Christos Zarkos*, Nevena Stojkovic*, Varun Gohil, Christina Delimitrou***

**Hermes' SERenaDE: Hardware Acceleration of Serialization Frameworks** (YArch'25)

***Christos Zarkos, Nikita Lazarev, Shabnam Sheikhha, Christina Delimitrou***

## Awards and Honors

**Athans Memorial PhD Fellowship** Sept. 2025

For the third year of PhD studies at MIT

**Onassis Foundation PhD Scholarship** Sept. 2024 - Aug. 2027

For PhD studies at MIT

**Greek Institute of National Scholarships, Excellence Scholarship** Dec. 2023

Awarded from the Greek Institute of National Scholarships for being the valedictorian of the class of 2023 for the Computer Science Department of the University of Crete

**Paris Kanellakis Fellowship** Sept. 2023

For the first year of PhD studies at MIT

**Elisavet Karamintzou Scholarship** July 2023

Awarded from the University of Crete for being the valedictorian of the class of 2023 for the Computer Science Department

**Stelios Orphanoudakis Scholarship** Sept. of 2020, 2021, and 2022

Awarded from ICS-FORTH all three possible years during my Bachelor studies in the Computer Science Department of UoC (2019-2020, 2020-2021 and 2021-2022) for being among the top3 students of the class (first all 3 years)

## Projects

**SPARROW-SV (Bachelor's thesis)**

***Implementation of a SIMD Unit for AI Acceleration for a RISC-V Processor*** SystemVerilog, C

- Implemented a SIMD unit in SystemVerilog and integrated it into an industry proven RISC-V processor (Veer EH1 RISC-V Core by Western Digital)
- Operated Logical Synthesis, area and timing analysis of the design with EDA tools (DesignCompiler)

**Ka-chow**

***[6.5930] Hardware Architecture for Deep Learning Final Project*** Jupyter Notebook, Python

- 3-member group project
- Implemented a simulation of the performance of the Lightning photonic accelerator on CiMLoop

**AlphaCompiler**

***CS340 Programming Languages and Compilers University project*** C

- 3-member group project
- Implemented a Compiler and a Virtual Machine for a javascript-like language (written in C)

## Service

**DATE 23 Conference, D9 track** Fall 2022

***Paper Sub-reviewer***

## Skills

**Programming Languages, Technologies & Tools :** SystemVerilog, C, C++, VHDL, Python, Java, R, OpenMP, Vivado, EDA, Git, LaTeX, Linux

**Languages :** Greek (Native Language), English (C2), Spanish (B2)

## Areas of interest

Computer Architecture, Computer Systems, Cloud Computing, Hardware/Software Codesign, FPGAs, Embedded Systems